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Programmable Logic Controllers (PLCs): An Overview

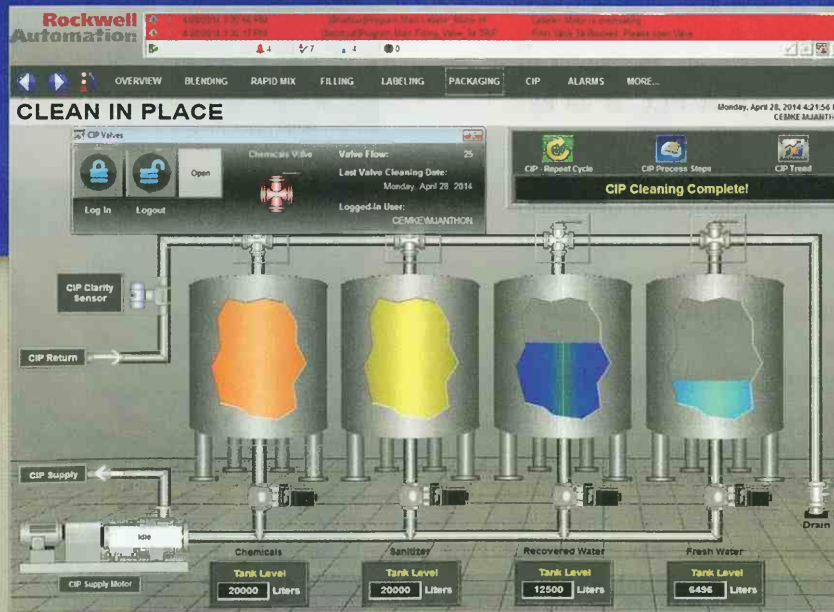


Image Courtesy of Rockwell Automation, Inc.

Chapter Objectives

After completing this chapter, you will be able to:

- Define what a programmable logic controller (PLC) is and list its advantages over relay systems
- Identify the main parts of a PLC and describe their functions
- Outline the basic sequence of operation for a PLC
- Identify the general classifications of PLCs

This chapter gives a brief history of the evolution of the programmable logic controller, or PLC. The reasons for changing from relay control systems to PLCs are discussed. You will learn the basic parts of a PLC, how a PLC is used to control a process, and the different kinds of PLCs and their applications. The ladder logic language, which was developed to simplify the task of programming PLCs, is introduced.

1.1 Programmable Logic Controllers

Programmable logic controllers are now the most widely used industrial process control technology. A **programmable logic controller (PLC)** is an industrial grade computer that is capable of being programmed to perform control functions. The programmable controller has eliminated much of the hardwiring associated with conventional relay control circuits. Other benefits include fast response, easy programming and installation, high control speed, network compatibility, troubleshooting and testing convenience, and high reliability.

The PLC is designed for multiple input and output arrangements, extended temperature ranges, immunity to electrical noise, and resistance to vibration and impact. Programs for the control and operation of manufacturing process equipment and machinery are typically stored in battery-backed or nonvolatile memory. A PLC is an example of a **real-time system** since the output of the system controlled by the PLC depends on the input conditions.

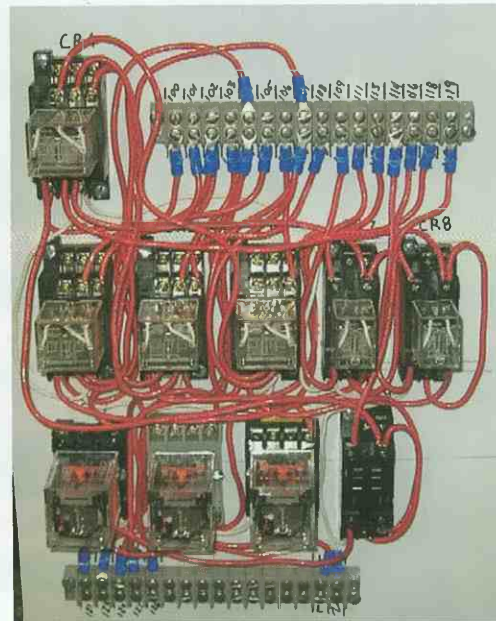
The PLC is, then, basically a digital computer designed for use in machine control. Unlike a personal computer, it has been designed to operate in the industrial environment and is equipped with special input/output interfaces and a control programming language.

Initially the PLC was used to replace **relay logic**, but its ever-increasing range of functions means that it is found in many and more complex applications. Because the structure of a PLC is based on the same principles as those employed in computer architecture, it is capable not only of performing relay switching tasks but also of performing other applications such as timing, counting, calculating, comparing, and the processing of analog signals.

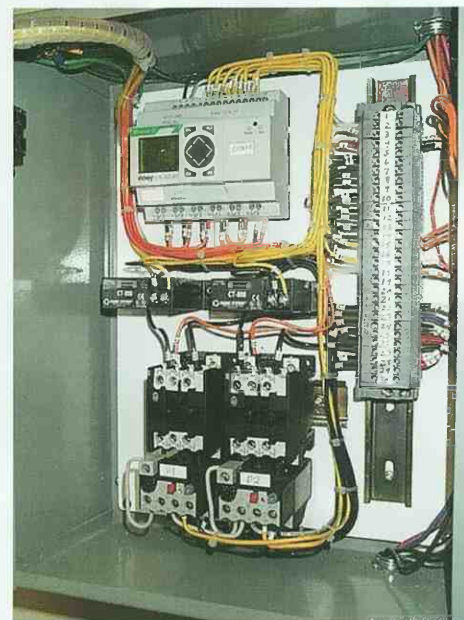
Programmable controllers offer several advantages over a conventional relay type of control. Relays have to be hardwired to perform a specific function. When the system requirements change, the relay wiring has to be changed or modified. In extreme cases, such as in the auto industry, complete control panels had to be replaced since it was not economically feasible to rewire the old panels with each model changeover. The programmable controller has eliminated much of the hardwiring associated with conventional relay control circuits (Figure 1-1). It is small and inexpensive compared to equivalent relay-based process control systems. Modern control systems still include relays, but these are rarely used for logic.

PLCs provide many other benefits including:

- **Increased Reliability.** Once a program has been written and tested, it can be easily downloaded to other PLCs. Since all the logic is contained in the PLC's memory, there is no chance of making a logic wiring error (Figure 1-2). The program takes the



(a)



(b)

Figure 1-1 Relay- and PLC-based control panels. (a) Relay-based control panel. (b) PLC-based control panel.

Source: (a) Courtesy Mid-Illini Technical Group, Inc.; (b) Photo courtesy of Ramco Electric Ltd., Toronto

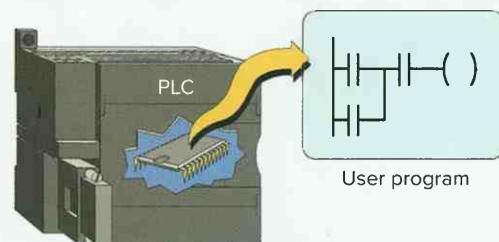


Figure 1-2 All the logic is contained in the PLC's memory.

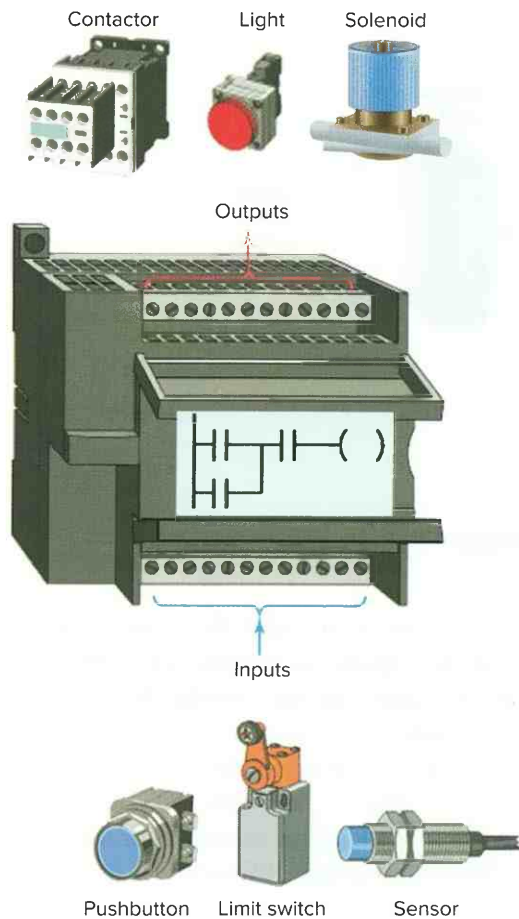


Figure 1-3 Relationships between the inputs and outputs are determined by the user program.

place of much of the external wiring that would normally be required for control of a process. Hardwiring, though still required to connect field devices, is less intensive. PLCs also offer the reliability associated with solid-state components.

- **More Flexibility.** It is easier to create and change a program in a PLC than to wire and rewire a circuit. With a PLC the relationships between the inputs and outputs are determined by the user program instead of the manner in which they are interconnected (Figure 1-3). Original equipment manufacturers can provide system updates by simply sending out a new program. End users can modify the program in the field, or if desired, security can be provided by hardware features such as key locks and by software passwords.
- **Lower Cost.** PLCs were originally designed to replace relay control logic, and the cost savings have been so significant that relay control is becoming obsolete except for power applications. Generally, if an application has more than about a half-dozen



Figure 1-4 High-speed counting.
Source: Courtesy of Banner Engineering Corp.

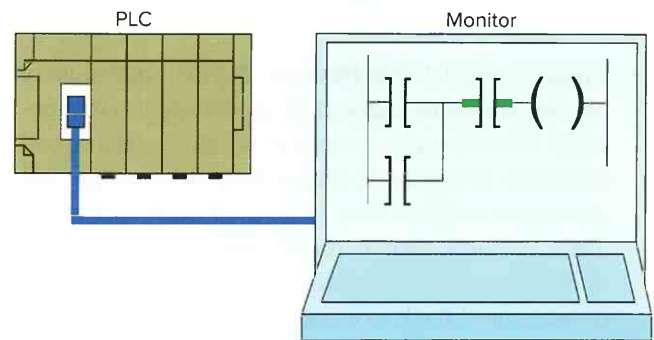


Figure 1-5 Control program can be displayed on a monitor in real time.

control relays, it will probably be less expensive to install a PLC.

- **Communications Capability.** A PLC can communicate with other controllers or computer equipment to perform such functions as supervisory control, data gathering, monitoring devices and process parameters, and download and upload of programs.
- **Faster Response Time.** PLCs are designed for high-speed and real-time applications (Figure 1-4). The programmable controller operates in real time, which means that an event taking place in the field will result in the execution of an operation or output. Machines that process thousands of items per second and objects that spend only a fraction of a second in front of a sensor require the PLC's quick-response capability.
- **Easier to Troubleshoot.** PLCs have resident diagnostics and override functions that allow users to easily trace and correct software and hardware problems. To find and fix problems, users can display the control program on a monitor and watch it in real time as it executes (Figure 1-5).

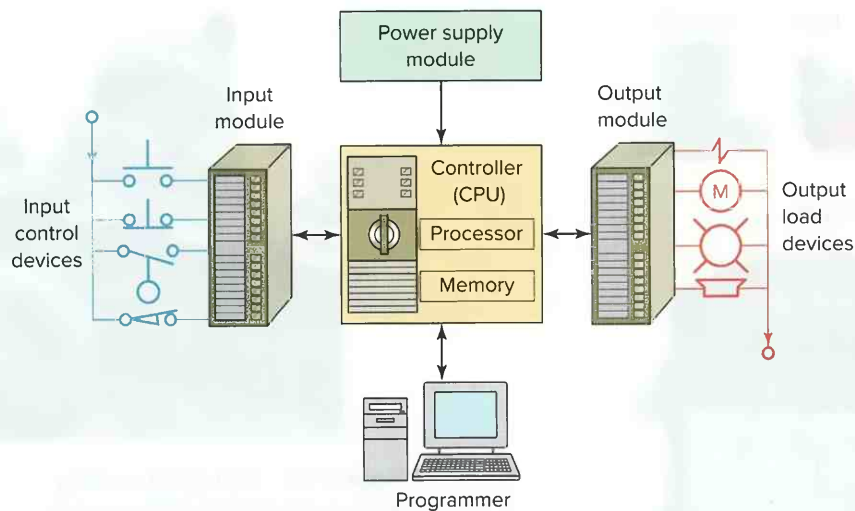


Figure 1-6 Typical parts of a programmable logic controller.

- **Easier to Test Field Devices.** A PLC control panel has the ability to check field devices at a common point. For example, a control system consisting of hundreds of input and output field devices may be contained within a very large manufacturing area. Thus, it would take a considerable amount of time to check each device at its location. By having each device wired back to a common point on a PLC module, each device could be checked for operation fairly quickly.

1.2 Parts of a PLC

A typical PLC can be divided into parts, as illustrated in Figure 1-6. These are the *central processing unit (CPU)*, the *input/output (I/O)* section, the *power supply*, and the *programming device*. The term *architecture* can refer to PLC hardware, to PLC software, or to a combination of both. An *open* architecture design allows the system to be connected easily to devices and programs made by other manufacturers. Open architectures use off-the-shelf components that conform to approved standards. A system with a *closed* architecture is one whose design is proprietary, making it more difficult to connect to other systems. Most PLC systems are in fact **proprietary**, so you must be sure that any generic hardware or software you may use is compatible with your particular PLC. Also, although the principal concepts are the same in all methods of programming, there are differences in addressing, memory allocation, retrieval, and data handling for different models. Consequently, PLC programs cannot be interchanged among different PLC manufacturers.

There are two ways in which I/Os (Inputs/Outputs) are incorporated into the PLC: fixed and modular. **Fixed I/O**

(Figure 1-7) is typical of small PLCs that come in one package with no separate, removable units. The processor and I/O are packaged together, and the I/O terminals will have a fixed number of connections built in for inputs and outputs. The main advantage of this type of packaging is lower cost. The number of available I/O points varies and usually can be expanded by buying additional units of fixed I/O. One disadvantage of fixed I/O is its lack of flexibility; you are limited in what you can get in the quantities and types dictated by the packaging. Also, for some models, if any part in the unit fails, the whole unit has to be replaced.

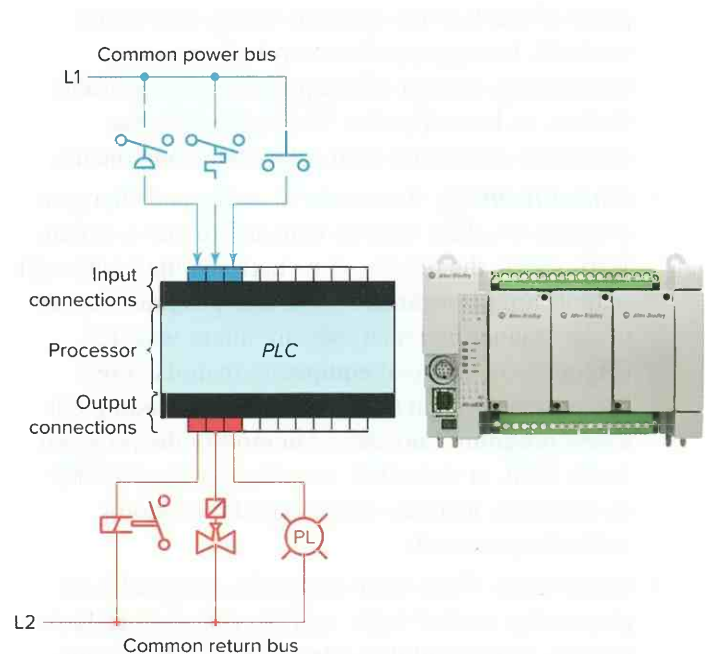


Figure 1-7 Fixed I/O configuration.

Source: Image Courtesy of Rockwell Automation, Inc

Modular I/O (Figure 1-8) is divided by compartments into which separate modules can be plugged. This feature greatly increases your options and the unit's flexibility. You can choose from the modules available from the manufacturer and mix them any way you desire. The basic modular controller consists of a rack, power supply, processor module (CPU), input/output (I/O modules), and an operator interface for programming and monitoring. The modules plug into a rack. When a module is slid into the rack, it makes an electrical connection with a series of contacts called the backplane, located at the rear of the rack. Communication between modules is accomplished by the backplane rail that each module plugs into.

The **power supply** supplies DC power through the backplane, to the processor and the other modules that plug into the rack (Figure 1-9). For large PLC systems, this power supply does not normally supply power to the field

devices. With most systems, power to field devices is provided by external alternating current (AC) or direct current (DC) supplies. For some small micro PLC systems, the power supply may be used to power field devices.

The **processor (CPU)** is the "brain" of the PLC. A typical processor usually consists of a microprocessor for implementing the logic and controlling the communications among the modules. The processor requires memory for storing user program instructions, numerical values, and I/O devices status.

The CPU controls all PLC activity and is responsible for running the program. The PLC program is executed as part of a repetitive process referred to as a scan (Figure 1-10). A typical PLC scan starts with the CPU reading the status of inputs. Then, the program logic is executed. Once the

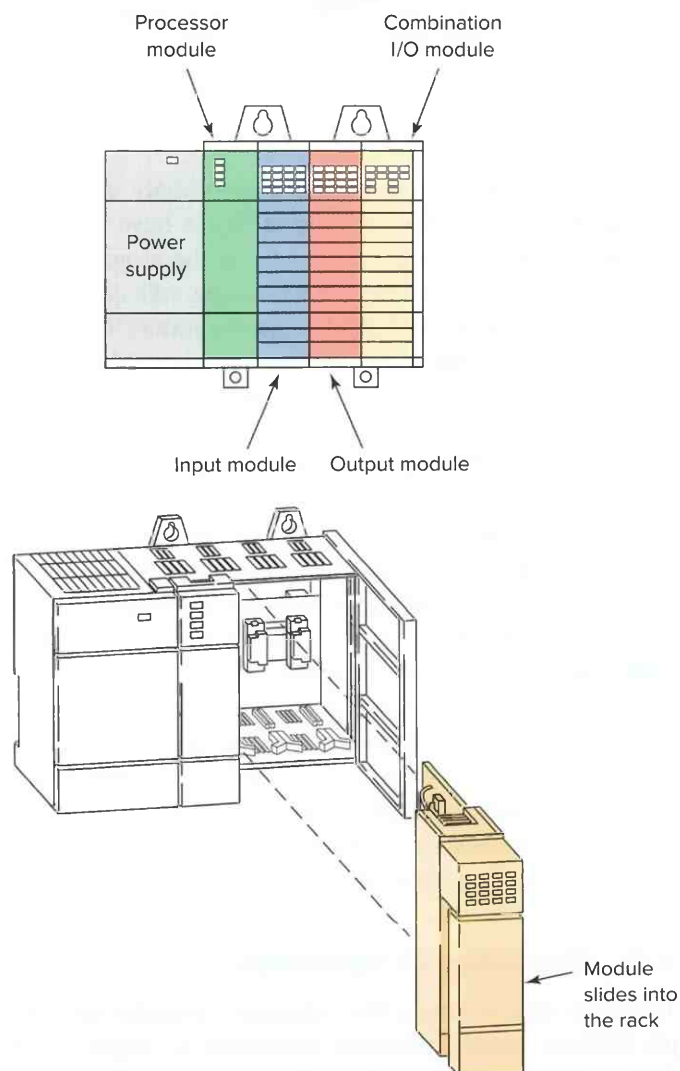


Figure 1-8 Modular I/O configuration.



Figure 1-9 The power supply supplies DC power to other modules that plug into the rack.

Source: Photo of PLC Modicon M340 © Schneider Electric, 2010.
www.schneider-electric.com.

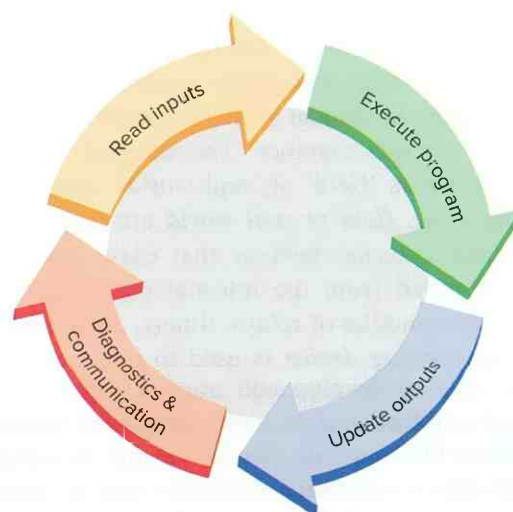


Figure 1-10 Typical PLC scan cycle.

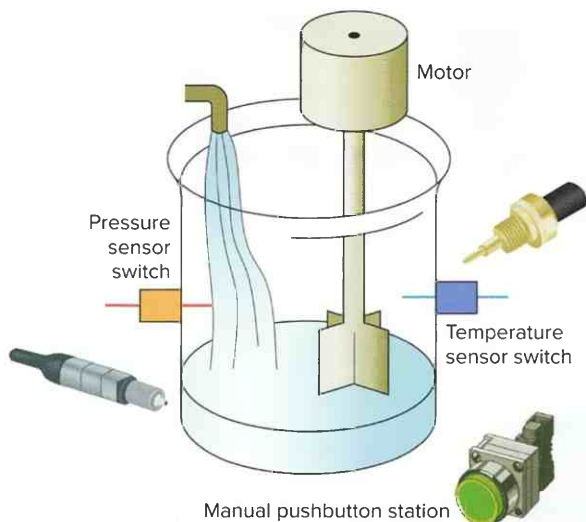


Figure 1-12 Mixer process control problem.

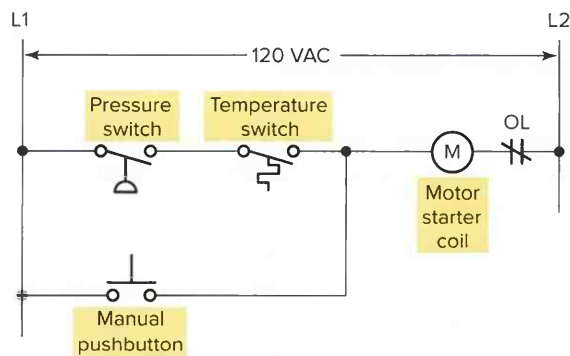


Figure 1-13 Process control relay ladder diagram.

preset values. In addition, direct manual operation of the motor is provided by means of a separate pushbutton station. The process is monitored with temperature and pressure sensor switches that close their respective contacts when conditions reach their preset values.

This control problem can be solved using the relay method for motor control shown in the relay ladder diagram of Figure 1-13. The motor starter coil (M) is energized when both the pressure and temperature switches are closed or when the manual pushbutton is pressed.

Now let's look at how a programmable logic controller might be used for this application. The same input field devices (pressure switch, temperature switch, and pushbutton) are used. Each of these devices is wired to a terminal on the 120 VAC input module as shown in Figure 1-14. The processor memory location addresses used are:

- I/1 for the Pressure switch
- I/2 for the Temperature switch
- I/3 for the Manual pushbutton

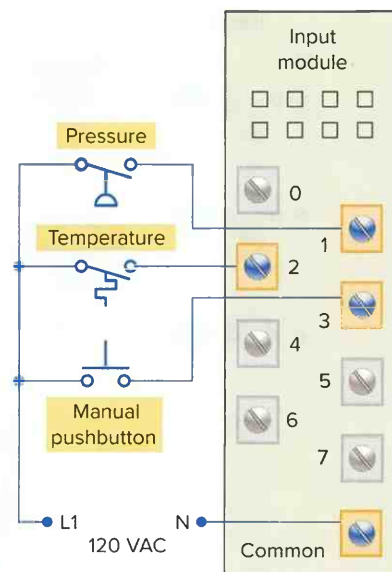


Figure 1-14 Typical wiring connections for a 120 VAC modular configured input module.

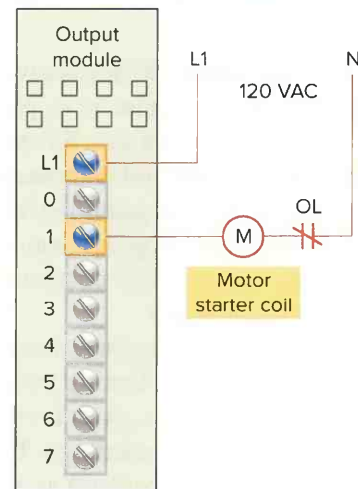


Figure 1-15 Typical wiring connections for a 120 VAC modular configured output module.

The same output field device (motor starter coil) would also be used. This device is wired to a terminal on the 120 VAC output module as shown in Figure 1-15. The processor memory location address used for the Motor starter coil is O/1.

Next, the PLC ladder logic program would be constructed and entered into the memory of the CPU. A typical ladder logic program for this process is shown in Figure 1-16. The format used is similar to the layout of the hardwired relay ladder circuit. The individual symbols represent instructions, whereas the numbers represent the instruction memory location addresses. To program the controller, you enter these instructions one by one into the processor memory from the programming

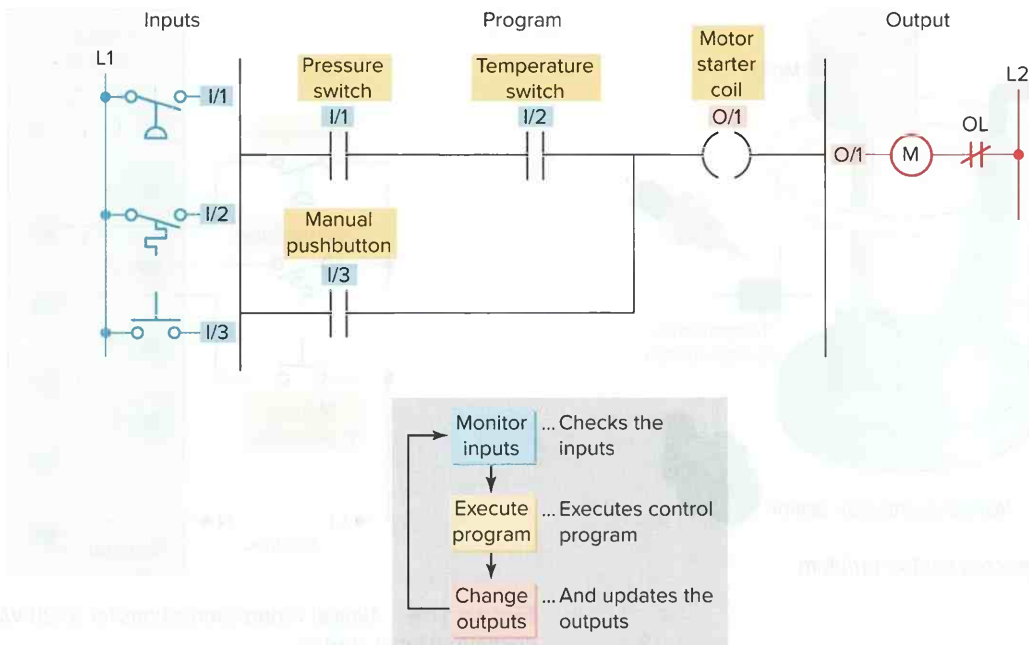


Figure 1-16 Process control PLC ladder logic program with typical addressing scheme.

device. Each input and output device is given an address, which lets the PLC know where it is physically connected. Note that the I/O address format will differ, depending on the PLC model and manufacturer. Instructions are stored in the user program portion of the processor memory. During the program scan the controller monitors the inputs, executes the control program, and changes the output accordingly.

For the program to operate, the controller is placed in the RUN mode, or operating cycle. During the program scan, the controller monitors the inputs, executes the control program, and changes the output accordingly. Each $\|$ symbol (looks like a normally open contact) is an instruction. The $()$ symbol is considered to represent a coil that, when energized, will energize the device that is wired to the respective output. In the ladder logic program of Figure 1-16, the coil O/1 is energized when contacts I/1 and I/2 are closed or when contact I/3 is closed. Either of these conditions provides a continuous logic path from left to right across the rung that includes the coil.

A programmable logic controller operates in real time in that an event taking place in the field will result in an operation or output taking place. The RUN operation for the process control logic can be described by the following sequence of events:

- First, the pressure switch, temperature switch, and pushbutton inputs are examined and their status is recorded in the controller's memory.
- A closed contact is recorded in memory as logic 1 and an open contact as logic 0.

- Next the ladder diagram is evaluated, with each internal contact given an OPEN or CLOSED status according to its recorded 1 or 0 state.
- When the states of the input contacts provide logic continuity from left to right across the rung, the output coil memory location is given a logic 1 value and the output module interface contacts will close.
- When there is no logic continuity of the program rung, the output coil memory location is set to logic 0 and the output module interface contacts will be open.
- The completion of one cycle of this sequence by the controller is called a *scan*. The scan time, the time required for one full cycle, provides a measure of the speed of response of the PLC.
- Generally, the output memory location is updated during the scan but the actual output is not updated until the end of the program scan during the I/O scan.

Figure 1-17 shows the typical wiring required to implement the process control scheme using a fixed PLC controller. In this example, the Allen-Bradley Pico controller equipped with 8 inputs and 4 outputs is used to control and monitor the process. Installation can be summarized as follows:

- Fused power lines, of the specified voltage type and level, are connected to the controller's L1 and L2 terminals.
- The pressure switch, temperature switch, and pushbutton field input devices are hardwired between L1

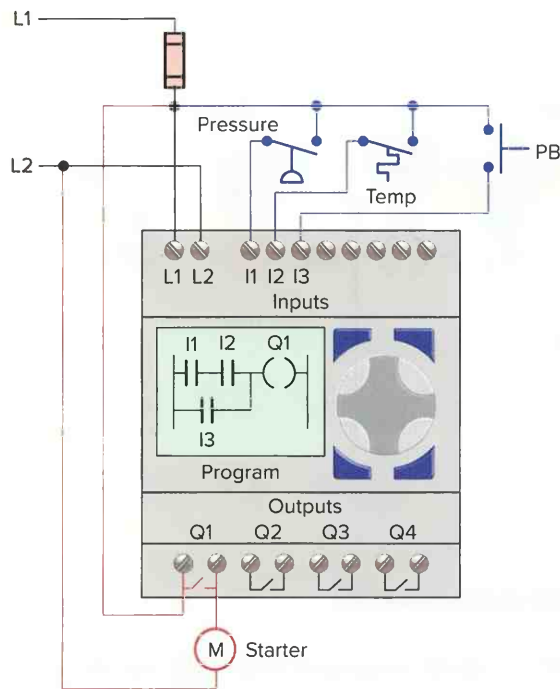


Figure 1-17 Typical wiring required to implement the process control scheme using a fixed PLC controller.

and controller input terminals I1, I2, and I3, respectively.

- The motor starter coil connects directly to L2 and in series with Q1 relay output contacts to L1.
- The ladder logic program is entered using the front keypad and LCD display.
- Pico programming software is also available that allows you to create as well as test your program using a personal computer.

1.4 Modifying the Operation

One of the important features of a PLC is the ease with which the program can be changed. For example, assume that the original process control circuit for the mixing operation must be modified as shown in the relay ladder diagram of Figure 1-18. The change requires that the manual pushbutton control be permitted to operate at any pressure, but not unless the specified temperature setting has been reached.

If a relay system were used, it would require some rewiring of the circuit shown in Figure 1-18 to achieve the desired change. However, if a PLC system were used, no rewiring would be necessary. The inputs and outputs are still the same. All that is required is to change the PLC ladder logic program as shown in Figure 1-19.

At times, a process may call for additional real input or output field devices to be added to the circuit. The

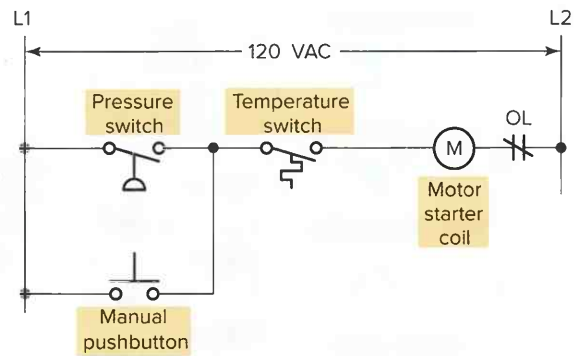


Figure 1-18 Relay ladder diagram for the modified process.

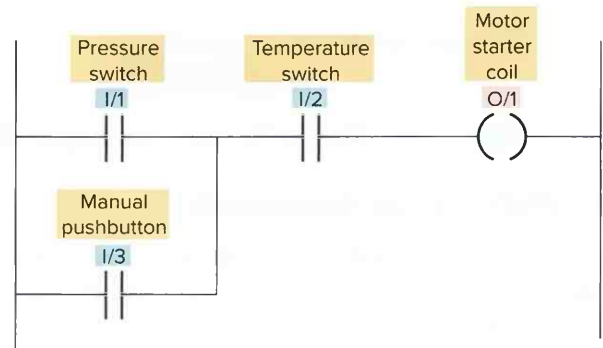


Figure 1-19 PLC ladder logic program for the modified process.

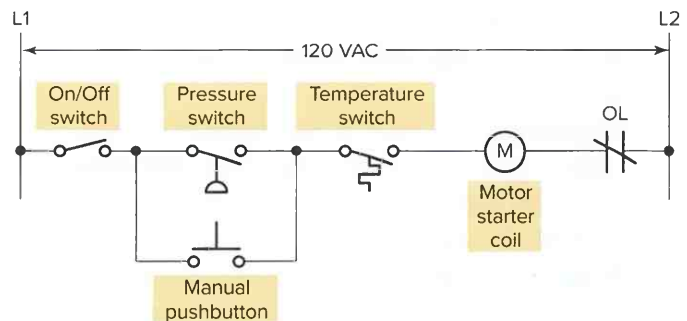


Figure 1-20 Modified relay ladder for the addition of an ON/OFF switch.

relay ladder diagram of Figure 1-20 shows the circuit further modified to include a process ON/OFF switch. To accomplish this, using hard-wired circuit control requires accessing both the pressure switch and manual pushbutton and rewiring the circuit so that the two are in parallel with each other and in series with the ON/OFF switch. The modification implemented using a PLC ladder logic program is shown in Figure 1-21. Note that the original wiring of the existing PLC input circuit remains the same. All that is required is the connection of the ON/OFF switch to I/4 input and the related change in the PLC program.

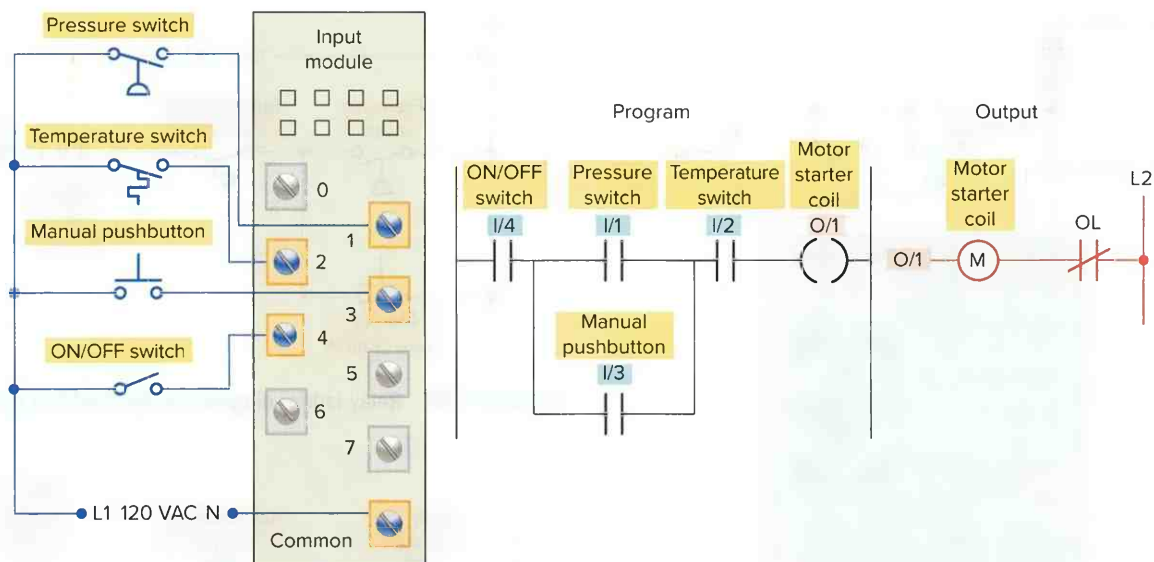


Figure 1-21 Modified PLC ladder logic program and ON/OFF switch module connection.

1.5 PLCs versus Computers

The architecture of a PLC is basically the same as that of a personal computer (PC). Characteristics that distinguish PLCs from PC include:

- **Hardware.** Unlike PCs, the PLC has no permanently attached keyboard, CD drive, or monitor.
- **Operating Environment.** PLCs are designed to operate in the industrial environment a wide range of ambient temperature, humidity, and electrical noise factors.
- **Programming.** The PLC is programmed in relay ladder logic or four other types of programming languages and comes with its program language built into its memory.
- **Program Execution.** PLCs execute a single program in a sequential order, while computers execute several programs or tasks simultaneously in any order.

Software associated with a PLC but written and run on a personal computer falls into the following two broad categories:

- PLC software that allows the user to program and document gives the user the tools to write a PLC program—using ladder logic or another programming language—and document or explain the program in as much detail as is necessary.
- PLC software that allows the user to monitor and control the process is called a **human machine interface (HMI)**. It enables the user to view a process—or a graphical representation of a process—on a monitor, operate the machine, trend values, and receive alarm conditions (Figure 1-22).

PLCs can be integrated with HMIs but the same software does not program both devices.

Most recently automation manufacturers have responded to the increased requirements of industrial control systems by blending the advantages of PLC-style control with that of PC-based systems. Such a device has been termed a *programmable automation controller*, or *PAC*. Programmable automation controllers combine PLC ruggedness with PC functionality. Using PACs, you can build advanced systems incorporating software capabilities such as advanced control, communication, data logging, and signal processing with rugged hardware performing logic, motion, process control, and vision. One main difference between PLC- and PAC-based programs is how the program is executed. A PLC mixes scan-based and event-driven program execution, whereas PAC software is typically event-driven. The scan-based execution of a PLC



Figure 1-22 Human Machine Interface (HMI).
Source: Image Courtesy of Rockwell Automation, Inc.

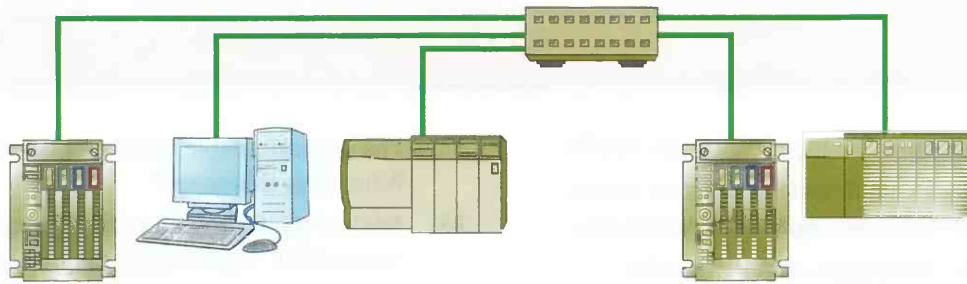


Figure 1-23 Control management PLC application.

program might take longer because the system needs to complete the higher priority actions in the cycle first.

1.6 PLC Size and Application

The criteria used in categorizing PLCs include functionality, number of inputs and outputs, cost, and physical size. Of these, the **I/O count** is the most important factor.

There are three major types of PLC application: single-ended, multitask, and control management. A **single-ended** or **stand-alone** PLC application involves one PLC controlling one process. This would be a stand-alone unit and would not be used for communicating with other computers or PLCs.

A **multitask** PLC application involves one PLC controlling several processes. Adequate I/O capacity is a significant factor in this type of installation. In addition, if the PLC would be a subsystem of a larger process and would have to communicate with a central PLC or computer, provisions for a data communications network are also required.

A **control management** PLC application involves one PLC controlling several others (Figure 1-23). This kind of application requires a large PLC processor designed to communicate with other PLCs and possibly with a computer. The control management PLC supervises several PLCs by downloading programs that tell the other PLCs what has to be done. It must be capable of connection to

all PLCs so that by proper addressing it can communicate with any one it wishes to.

Memory is the part of a PLC that stores data, instructions, and the control program. Memory size is usually expressed in K values: 1 K, 6 K, 12 K, and so on. The measurement kilo, abbreviated K, normally refers to 1000 units. When dealing with computer or PLC memory, however, 1 K means 1024, because this measurement is based on the binary number system ($2^{10} = 1024$).

Although it is common for us to measure the memory capacity of PLCs in words, we need to know the number of bits in each word before memory size can be accurately compared. Modern computers usually have a word size of 16, 32, or 64 bits. The amount of memory required depends on the application. Factors affecting the memory size needed for a particular PLC installation include:

- Number of I/O points used
- Size of control program
- Data-collecting requirements
- Supervisory functions required
- Future expansion

The **instruction set** for a particular PLC lists the different types of instructions supported. Typically, this ranges from 15 instructions on smaller units up to 100 instructions on larger, more powerful units (see Table 1-1).

Table 1-1 Typical PLC Instructions

Instruction	Operation
XIC (Examine ON).	Examine a bit for an ON or 1 condition
XIO (Examine OFF).	Examine a bit for an OFF or 0 condition
OTE (Output Energize).	Turn ON a bit (nonretentive)
OTL (Output Latch).	Latch a bit (retentive)
OTU (Output Unlatch)	Unlatch a bit (retentive)
TOF (Timer Off-Delay)	Turn an output ON or OFF after its rung has been OFF for a preset time interval
TON (Timer On-Delay).	Turn an output ON or OFF after its rung has been ON for a preset time interval
CTD (Count Down).	Use a software counter to count down from a specified value
CTU (Count Up)	Use a software counter to count up to a specified value



1. What is a programmable logic controller (PLC)?
2. Identify four tasks in addition to relay switching operations that PLCs are capable of performing.
3. List six distinct advantages that PLCs offer over conventional relay-based control systems.
4. Explain the differences between open and proprietary PLC architecture.
5. State two ways in which I/O is incorporated into the PLC.
6. Describe how the I/O modules connect to the processor in a modular-type PLC configuration.
7. Explain the main function of each of the following major components of a PLC:
 - a. Processor module (CPU)
 - b. I/O modules
 - c. Programming device
 - d. Power supply module
8. What are the two most common types of PLC programming devices?
9. Explain the terms *program* and *programming language* as they apply to a PLC.
10. What is the standard programming language used with PLCs?
11. Answer the following with reference to the process control relay ladder diagram of Figure 1-13 of this chapter:
 - a. When do the pressure switch contacts close?
 - b. When do the temperature switch contacts close?
 - c. How are the pressure and temperature switches connected with respect to each other?
 - d. Describe the two conditions under which the motor starter coil will become energized.
 - e. What is the approximate value of the voltage drop across each of the following when their contacts are open?
 - (1) Pressure switch
 - (2) Temperature switch
 - (3) Manual pushbutton
12. The programmable controller operates in real time. What does this mean?
13. Answer the following with reference to the process control PLC ladder logic diagram of Figure 1-16 of this chapter:
 - a. What do the individual symbols represent?
 - b. What do the numbers represent?
 - c. What field device is the number I/2 identified with?
 - d. What field device is the number O/1 identified with?
 - e. What two conditions will provide a continuous path from left to right across the rung?
 - f. Describe the sequence of operation of the controller for one scan of the program.
14. Compare the method by which the process control operation is changed in a relay-based system to the method used for a PLC-based system.
15. Compare the PLC and PC with regard to:
 - a. Physical hardware differences
 - b. Operating environment
 - c. Method of programming
 - d. Execution of program
16. What two categories of software written and run on PCs are used in conjunction with PLCs?
17. What is a programmable automation controller (PAC)?
18. List four criteria by which PLCs are categorized.
19. Compare the single-ended, multitask, and control management types of PLC applications.
20. What is the memory capacity, expressed in bits, for a PLC that uses 16-bit words and has an 8 K word capacity?
21. List five factors affecting the memory size needed for a particular PLC installation.
22. What does the instruction set for a particular PLC refer to?

1. Given two single-pole switches, write a program that will turn on an output when both switch *A* and switch *B* are closed.
2. Given two single-pole switches, write a program that will turn on an output when either switch *A* or switch *B* is closed.
3. Given four NO (Normally Open) pushbuttons (*A-B-C-D*), write a program that will turn a lamp on if pushbuttons *A* and *B* or *C* and *D* are closed.
4. Write a program for the relay ladder diagram shown in Figure 1-24.

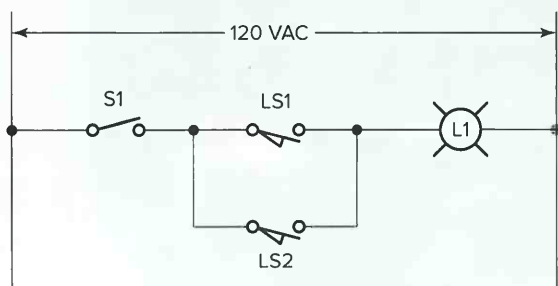


Figure 1-24 Circuit for Problem 4.

5. Write a program for the relay ladder diagram shown in Figure 1-25.

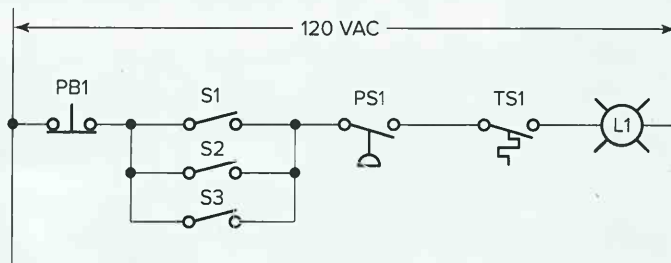


Figure 1-25 Circuit for Problem 5.

PLC Hardware Components



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This chapter exposes you to the details of PLC hardware and modules that make up a PLC control system. The chapter's illustrations show the various parts of a PLC as well as general connection paths. In this chapter we discuss the CPU and memory hardware components, including the various types of memory that are available, and we describe the hardware of the input/output section, including the difference between the discrete and analog types of modules.

Chapter Objectives

After completing this chapter, you will be able to:

- List and describe the function of the hardware components used in PLC systems
- Describe the basic circuitry and applications for discrete and analog I/O modules, and interpret typical I/O and CPU specifications
- Differentiate between tag-based and rack/slot-based addressing.
- Describe the general classes and types of PLC memory devices
- List and describe the different types of PLC peripheral support devices available